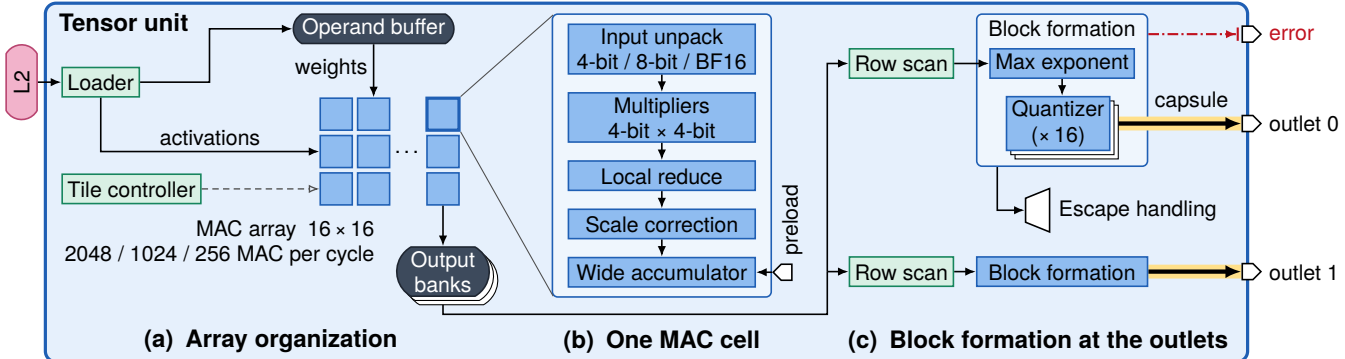
Tensor unit
L2
Loader
Operand buffer
weights
activations
Tile controller
MAC array 16 × 16
2048 / 1024 / 256 MAC per cycle
Output banks
(a) Array organization
Input unpack
4-bit / 8-bit / BF16
Multipliers
4-bit × 4-bit
Local reduce
Scale correction
Wide accumulator
preload
(b) One MAC cell
Row scan
Block formation
Max exponent
Quantizer
(× 16)
capsule
error
outlet 0
Escape handling
Row scan
Block formation
outlet 1
(c) Block formation at the outlets